

Math 113 – Abstract Algebra

Homework 3 — my solutions

Fall 2026 ■ Keshav Ramamurthy

All assignments must be **turned in on Gradescope** by the deadline. Late assignments are not accepted.

Have questions? Work with your peers, attend office hours, check out the [Student Learning Center](#), and ask in discussion section!

Problems for credit. Turn in these problems; some subset of them will be graded.

1. Fraleigh Exercise 6.6
2. Fraleigh Exercise 6.14
3. Fraleigh Exercise 6.34
4. Fraleigh Exercise 6.44
5. Fraleigh Exercise 7.4
6. Fraleigh Exercise 7.8
7. Fraleigh Exercise 7.19

Practice problems. The following problems will not be graded, and should not be turned in, but are recommended as good practice problems. Solutions to most odd-numbered exercises are available in the back of the textbook.

- Fraleigh Exercises 6.3, 6.7, 6.9, 6.13, 6.17, 6.33, 6.35, 6.45, 6.47, 6.49, 6.51
- Fraleigh Exercises 7.1, 7.3, 7.7, 7.11, 7.13, 7.15, 7.17

Solutions

Problem 1 (Fraleigh Exercise 6.6)*Solution.*

□

Problem 2 (Fraleigh Exercise 6.14)*Solution.*

□

Problem 3 (Fraleigh Exercise 6.34)*Solution.*

□

Problem 4 (Fraleigh Exercise 6.44)*Solution.*

□

Problem 5 (Fraleigh Exercise 7.4)*Solution.*

□

Problem 6 (Fraleigh Exercise 7.8)*Solution.*

□

Problem 7 (Fraleigh Exercise 7.19)*Solution.*

□